CHE 201 Fall 2019 Quiz 2

Ganesh Shetty

TOTAL POINTS

100 / 100

QUESTION 1

1 85 pts

1.1 a 40 / 40

✓ - **0 pts** Correct

- **5 pts** h2 is not calculated or wrong
- **10 pts** KE is not calculated
- **5 pts** h1 is wrong
- **10 pts** PE not calculated

1.2 b+c 45 / 45

✓ - **0 pts** Correct

- **45 pts** not done or wrong
- **5 pts** unit conversion
- **10 pts** b is wrong

QUESTION 2

2 2 15 / 15

✓ - **0 pts** Correct

- **15 pts** not done

CHE 201 INTRO TO THERMODYNAMICS **October 25th, 2019**

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QUIZ 2: (100 pts) (Close book/closed notes) Show all your work for full credit

1.(85 points) The power output of steam turbine is 5MW, and the inlet and the exit conditions are as follow: P_1=2MPa, T_1=400^{0}C, V_1=50m/s, z_1= 10m; P_2=15kPa, x_2=0.90, V_2=180m/s, z_2=6m. a)Compare the magnitudes of delta h, delta KE, delta PE. b) Determine the work done per unit mass of the steam flowing through the turbine. c) Calculate the mass flow rate of the steam.

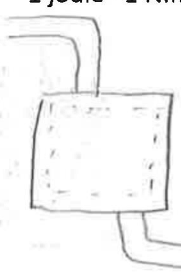
2. (15 points) Briefly explain the process that Lanzatech patented. (Monday's guest speaker; Rachel Brenc)

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e\right)$$

$$\dot{m} = \frac{AV}{v}$$

1 joule= 1 N.m 1 N= 1 kg.m/s^2

1)



steady state
one inlet one outlet

$\frac{dE_{cv}}{dt} = 0 = \frac{dm_{cv}}{dt}$

$\dot{m}_i = \dot{m}_e = \dot{m}$

a) $h_1 = 3247.6$ kJ/kg

$P_2 = 0.15$ bar

h_f at 0.15 bar ⇒ $\frac{251.4 - h_f}{251.4 - 191.83} = \frac{0.2 - 0.15}{0.2 - 0.1}$

$h_f = 221.615$ kJ/kg

h_g at 0.15 bar ⇒ $\frac{2609.7 - h_g}{2609.7 - 2584.7} = \frac{0.2 - 0.15}{0.2 - 0.1}$

$h_g = 2597.2$ kJ/kg

$h_e = 0.1 \times 221.615 + 0.9 \times 2597.2$ kJ/kg

$h_e = 2359.6$ kJ/kg

$\Delta h = h_i - h_e = 887.96$ kJ/kg not negligible

$V_1 = 50$ m/s $V_2 = 180$ m/s

$\frac{V_1^2 - V_2^2}{2000} = -14.95$ kJ/kg not negligible

$\Delta PE = g(z_1 - z_2)$

$= 9.81(10 - 6)$

$= 0.03924 \text{ kJ/kg}$

can be neglected.

b. $\left[(h_i - h_e) + \left(\frac{v_1^2 - v_2^2}{2}\right) + g(z_1 - z_2)\right]$

$= 873.05 \text{ kJ/kg}$

c. $\dot{W}_{cv} = \dot{m}\left[(h_i - h_e) + \left(\frac{v_1^2 - v_2^2}{2}\right) + g(z_1 - z_2)\right]$

$\dot{m} = \frac{5 \times 10^3 \text{ kW}}{873.05 \text{ kJ/kg}} = 5.73 \text{ kg/s}$

$\boxed{2}$ Lanzatech is a pro-environment company. It is a new start-up compared to other bigger companies like Exxon & BP. They work to change the harmful gases given off by factories. They convert it into less harmfull gas by the action of microbes onto the more harmful gases. These microbes behave like filtration system. Lanzatech is also working to alter the genes of the microbes to make them more efficient. Lanzatech has collabration with many international companies like BASF and Indian Oil.